



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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SATHYAMANGALAM - 638401 ERODE DISTRICT TAMILNADU INDIA
Ph : 04295-226000/221289 Fax : 04295-226666 E-mail : stayahead@bitsathy.ac.in Web : www.bitsathy.ac.in

22GE004 - BASICS OF ELECTRONICS ENGINEERING

UNIT IV & LOGIC SYNTHESIS USING DIODE AND TRANSISTORS

1. LOGIC GATES

Logic gates are an important concept if you are studying electronics. These are important digital devices that are mainly based on the Boolean function. Logic gates are used to carry out logical operations on single or multiple binary inputs and give one binary output. In simple terms, logic gates are the electronic circuits in a digital system.

1.1. Types of Basic Logic Gates

There are several basic logic gates used in performing operations in digital systems. The common ones are

- OR Gate
- AND Gate
- NOT Gate
- XOR Gate

Additionally, these gates can also be found in a combination of one or two. Therefore, we get other gates, such as NAND Gate, NOR Gate, XOR Gate and XNOR Gate.

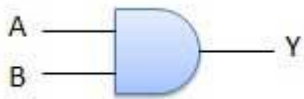
1.1.1. OR Gate

Logic gates are the basic building blocks of any digital system. It is an electronic circuit having one or more than one input and only one output. The relationship between the input and the output is based on a certain logic. Based on this, logic gates are named as AND gate, OR gate, NOT gate etc.

1.1.2. AND Gate

A circuit which performs an AND operation is shown in figure. It has n input ($n \geq 2$) and one output.

Logic diagram



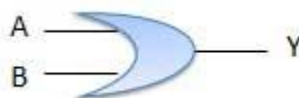
Truth Table

Inputs		Output
A	B	AB
0	0	0
0	1	0
1	0	0
1	1	1

1.1.3. OR Gate

A circuit which performs an OR operation is shown in figure. It has n input ($n \geq 2$) and one output.

Logic diagram



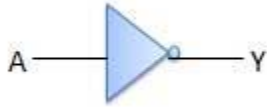
Truth Table

Inputs		Output
A	B	A + B
0	0	0
0	1	1
1	0	1
1	1	1

1.1.4 NOT Gate

NOT gate is also known as Inverter. It has one input A and one output Y.

Logic diagram



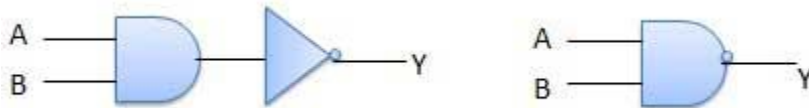
Truth Table

Inputs		Output
A		B
0		1
1		0

1.1.5. NAND Gate

A NOT-AND operation is known as NAND operation. It has n input ($n \geq 2$) and one output.

Logic diagram



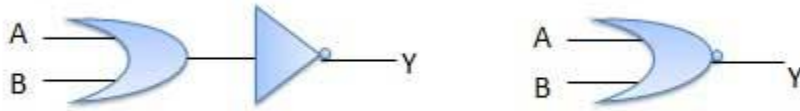
Truth Table

Inputs		Output
A	B	$\overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

1.1.6. NOR Gate

A NOT-OR operation is known as NOR operation. It has n input ($n \geq 2$) and one output.

Logic diagram



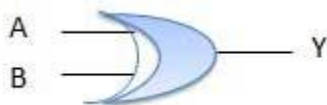
Truth Table

Inputs		Output
A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

1.1.7. XOR Gate

XOR or Ex-OR gate is a special type of gate. It can be used in the half adder, full adder and subtractor. The exclusive-OR gate is abbreviated as EX-OR gate or sometime as X-OR gate. It has n input ($n \geq 2$) and one output.

Logic diagram



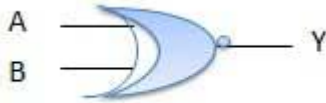
Truth Table

Inputs		Output
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

1.1.8. XNOR Gate

XNOR gate is a special type of gate. It can be used in the half adder, full adder and subtractor. The exclusive-NOR gate is abbreviated as EX-NOR gate or sometime as X-NOR gate. It has n input ($n \geq 2$) and one output.

Logic diagram



Truth Table

Inputs		Output
A	B	$A \oplus B$
0	0	1
0	1	0
1	0	0
1	1	1

2. PN Junction Diodes and BJT Transistors as Switch

Diode is a two terminal PN junction that can be used in various applications. One of such applications is an electrical switch. The PN junction, when forward biased acts as close circuited and when reverse biased acts as open circuited. Hence the change of forward and reverse biased states makes the diode work as a switch, the forward being ON and the reverse being OFF state.

2.1. Working of Diode as a Switch

Whenever a specified voltage is exceeded, the diode resistance gets increased, making the diode reverse biased and it acts as an open switch. Whenever the voltage applied is below the reference voltage, the diode resistance gets decreased, making the diode forward biased, and it acts as a closed switch. The following circuit explains the diode acting as a switch.

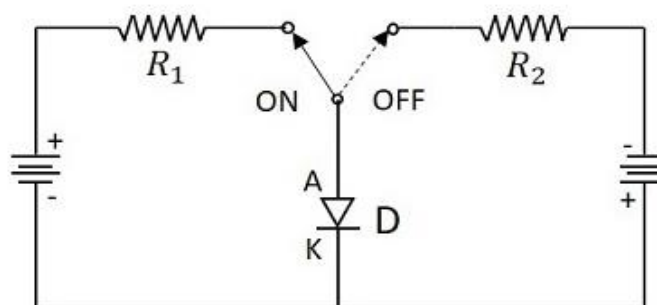


Figure.2.1 Switching circuit using Diode

2.2 Factors that affect diode switching times

There are few factors that affect the diode switching times, such as

- Diode Capacitance – The PN junction capacitance changes depending upon the bias conditions.
- Diode Resistance – The resistance offered by the diode to change its state.

- Doping Concentration – The level of doping of the diode affects the diode switching times.
- Depletion Width – The narrower the width of the depletion layer, the faster the switching will be. A Zener diode has a narrower depletion region than an avalanche diode, which makes the former a better switch.

2.3. Applications

There are many applications in which diode switching circuits are used, such as –

- High speed rectifying circuits
- High speed switching circuits
- RF receivers
- General purpose applications
- Consumer applications
- Automotive applications
- Telecom applications etc.

3. BJT Switching Operation

The operation of a BJT as a switching device is really easy to understand especially if you have already learned how to make a BJT operate in the cutoff and saturation region. A BJT acts like an open switch when it operates in the cutoff region. In figure 7, you can see that the npn BJT is operating in the cutoff region since the voltage at the base terminal is 0V. Therefore, the base-emitter junction is not forward-biased and if we will not consider the leakage current, all of the currents, I_B and I_C are zero. You can also see in figure 7 that the collector and emitter terminals act like an open SPST switch. Since I_C is zero, the voltage across the collector and emitter terminal, $V_{CE}(\text{cutoff})$, is equal to V_{CC} .

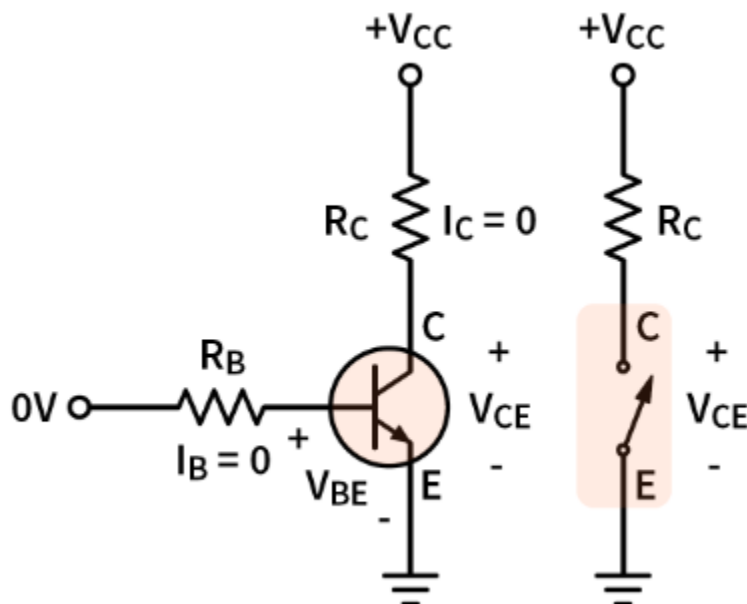


Figure 3.1. An NPN BJT operating in the cutoff region acts like an open switch.

So for a BJT to act as an open switch, all you need to do is to make sure that its base-emitter junction is not forward-biased. Now, for a BJT to act as a closed switch, it needs to operate in the saturation region. In figure 8, we've assumed that the NPN BJT is operating in the saturation region. As you can see, the collector and emitter terminals act like a closed SPST switch. Ideally, there's basically a short between the collector and emitter and the voltage drop across it should be zero. However, in reality, there is a small voltage drop across the collector and emitter terminals which is known as the saturation voltage, $V_{CE(sat)}$.

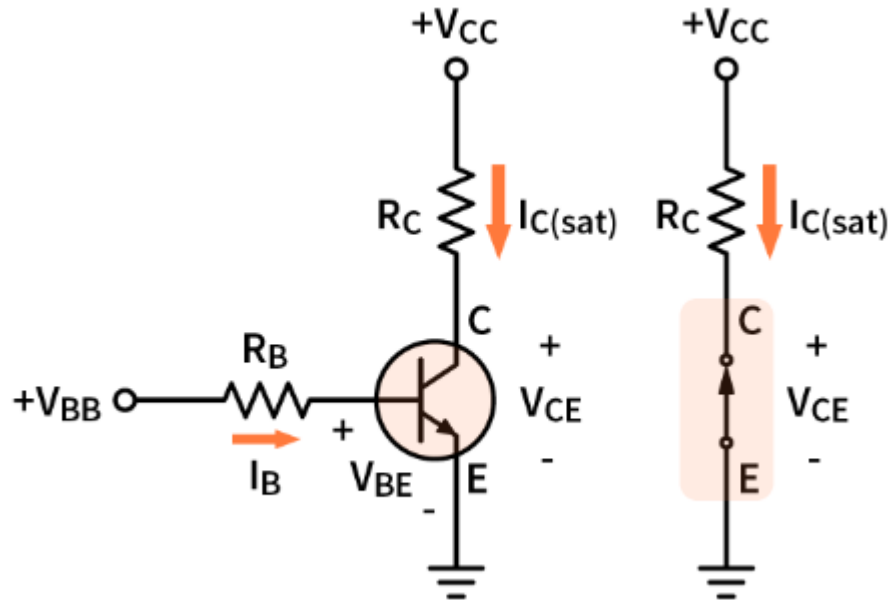


Figure 3.2. An npn BJT operating in the saturation region acts like a closed switch.

For a BJT to operate in the saturation region, the base-emitter junction and base-collector junction should be forward-biased, and there should be a sufficient base current to produce the collector saturation current, $I_{C(sat)}$. Using the circuit in figure 8, the formula to calculate the $I_{C(sat)}$ is:

$$I_{C(sat)} = \frac{V_{CC} - V_{CE(sat)}}{R_C}$$

$I_{C(sat)}$ is the current that the load R_C requires. $V_{CE(sat)}$ is specified in the datasheet of the BJT that you are going to use. After calculating $I_{C(sat)}$, the next thing to figure out is the minimum base current, $I_{B(min)}$, needed to make sure to produce $I_{C(sat)}$. You can use this equation to calculate $I_{B(min)}$.

$$I_{B(\min)} = \frac{I_{C(\text{sat})}}{\beta_{DC}}$$

To make sure that the BJT is operating in the saturation region, you just need to make sure that I_B is greater than $I_{B(\min)}$.